

MECALYS SA — Supplier Audit Report

AUDIT-SUP-2026-014 · TechAlu Foundry SAS (SUP-007)

Audit date: 2026-03-12 · Site: Saint-Priest foundry (FR) · Lead auditor: Supplier Quality Engineering · Audit type: for-cause process audit · Classification: internal

1 — Context and trigger

TechAlu Foundry SAS is the sole qualified source for the cast aluminium housings of the Vektor range (part AL-7043) and supplies five further housing and bracket references. This for-cause audit was triggered by three converging signals observed since January 2026: a rising number of porosity non-conformities at incoming inspection, a degradation of on-time delivery from 78% to 71% over the rolling three-month window, and a unit price increase of 18% on cast housing references notified on 2026-02-10 and effective 2026-03-01, raising AL-7043 from 39.66 EUR to 46.80 EUR. The supplier attributes the increase to aluminium ingot prices and energy costs.

2 — Findings

- F1 — Die temperature control drift (major).** The casting cell for the AL-70xx family operates without closed-loop die temperature regulation; thermocouple records show excursions beyond the qualified window on 9 of 20 sampled production days. This is the most probable root cause of the porosity defects observed at Mecalys incoming inspection.
- F2 — Delivery performance (major).** OTD measured at 71% over the rolling three-month window, below the 85% threshold of PROC-PUR-2025-11 §5.1. Root causes identified: a furnace rebuild consuming capacity, and order backlog prioritisation favouring larger automotive customers.
- F3 — Capacity and scheduling transparency (minor).** No forward capacity commitment is shared with Mecalys; confirmed due dates are systematically optimistic.
- F4 — Single-source exposure (observation).** The single-source waiver W-2024-031 covering class A part AL-7043 expires on 2026-05-31. No renewal file has been initiated and no second source qualification is in progress.

3 — Corrective action plan (CAP)

The following actions were agreed with the supplier's plant management. The overall CAP completion deadline is **2026-06-15**; a follow-up audit shall be scheduled within 30 days of completion.

#	Action	Owner	Due
A1	Install closed-loop die temperature regulation on the AL-70xx casting cell	TechAlu	2026-05-15
A2	100% X-ray inspection of AL-7043 batches until A1 is closed and verified	TechAlu	immediate
A3	Weekly delivery commitment review with Mecalys planning, with reports to Mecalys	TechAlu/Mecalys	2026-04-01
A4	Restore OTD to at least 85% on the rolling three-month window	TechAlu	2026-06-15
A5	Renew or replace single-source waiver W-2024-031; open second source qualification file	Mecalys/Supplier	2026-05-31

4 — Conclusion

The supplier remains qualified. Continuation of supply is conditioned on execution of the CAP by 2026-06-15. Should the CAP deadline lapse without closure, PROC-PUR-2025-11 §5.1 applies: new order placement shall be restricted and the situation escalated to the Supply Chain Director. Particular attention is drawn to finding F4: beyond 2026-05-31 and absent a renewed waiver, the single-sourcing of AL-7043 constitutes a standing non-compliance with §4.2 of the procedure.